Fractional-Charge Curvature and Band Gaps in Silicon and Diamond

A comparative PBEsol analysis

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October 14, 2026

Three quantities

Gap definitions

$$E_g^{KS} = E_{\mathrm{CBM}} - E_{\mathrm{VBM}}$$

$$E_g^{deriv} = \left.\frac{\partial E}{\partial N}\right|_{N^+} - \left.\frac{\partial E}{\partial N}\right|_{N^-}$$

$$E_g^{integer} = E(N-1) - 2E(N) + E(N+1)$$

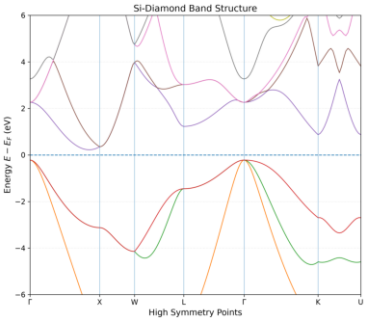
Computational design

	Si	Diamond
Structure	diamond	diamond
Atoms / electrons	2 / 8	2 / 8
Functional	PBEsol	PBEsol
Basis	DZP	DZP
k mesh	10^3	8^3
Mesh cutoff	200 Ry	400 Ry
a_0	5.4421 Å	3.5781 Å

Same structural family; very different electronic gap scales.

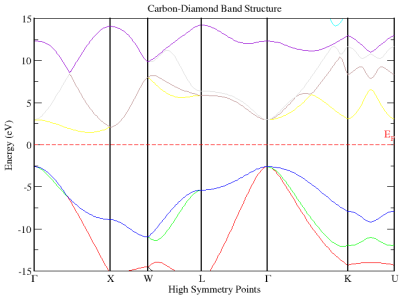
Kohn–Sham band structures

Silicon



$E_g^{KS} = \mathbf{0.45602719}$ eV, indirect

Diamond



$E_g^{KS} = \mathbf{4.0854}$ eV, indirect

Primitive Si: integer ΔSCF

For neutral primitive Si, $N = 8$.

State	Electrons	Total energy (eV)
$N-1$	7	-224.352300
N	8	-230.042602
$N+1$	9	-233.294319

$$I = E(7) - E(8) = 5.690302\ \mathrm{eV},$$

$$A = E(8) - E(9) = 3.251717\ \mathrm{eV}.$$

$$\begin{aligned} E_g^{\Delta\mathrm{SCF}} &= E(7) - 2E(8) + E(9) \\ &= -224.352300 - 2(-230.042602) - 233.294319 \\ &= \boxed{2.438585\ \mathrm{eV}}. \end{aligned}$$

$E_g^{KS} = 0.45602719\ \mathrm{eV}$ vs. $E_g^{\Delta\mathrm{SCF}} = 2.438585\ \mathrm{eV}$.

Primitive diamond: integer ΔSCF

For neutral primitive diamond, $N = 8$.

State	Electrons	Total energy (eV)
$N-1$	7	-318.806581
N	8	-326.444217
$N+1$	9	-325.884479

$$I = E(7) - E(8) = 7.637636\ \mathrm{eV},$$

$$A = E(8) - E(9) = -0.559738\ \mathrm{eV}.$$

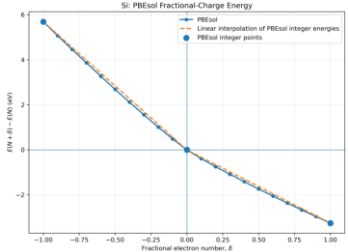
$$\begin{aligned} E_g^{\Delta\mathrm{SCF}} &= E(7) - 2E(8) + E(9) \\ &= -318.806581 - 2(-326.444217) - 325.884479 \\ &= \boxed{8.197374\ \mathrm{eV}}. \end{aligned}$$

$$E_g^{KS} = 4.0854\ \mathrm{eV} \qquad \text{vs.} \qquad E_g^{\Delta\mathrm{SCF}} = 8.197374\ \mathrm{eV}.$$

Here $A < 0$ because the calculated $N+1$ state is higher in total energy than the neutral state.

Fractional-charge curvature

Silicon



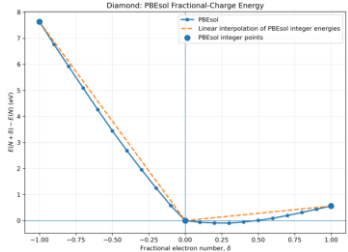
$$\boxed{E_{\mathrm{line}}(N_0+\delta) = (1-\delta)E(N_0) + \delta E(N_0+1)}$$

$$\Delta E(\delta) = E_{\mathrm{PBEsol}}(N+\delta) - E_{\mathrm{lin}}(N+\delta).$$

$$\Delta E(7.5) = -0.1582\ \mathrm{eV}, \qquad \Delta E(8.5) = -0.1140\ \mathrm{eV}.$$

$$\boxed{\Delta E < 0 \quad \Rightarrow \quad \text{convex } E(N)}$$

Diamond

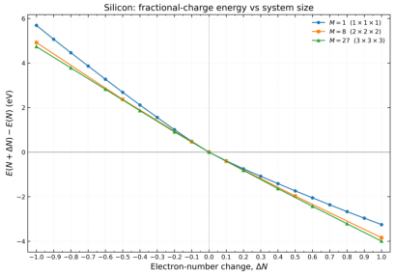


Diamond shows the same behavior:

$$\boxed{E_{\mathrm{PBEsol}} < E_{\mathrm{lin}}}$$

$$\boxed{\text{convex } E(N) \Longrightarrow \text{delocalization tendency}}$$

Si: why does the curve become straighter as the system grows?



$M = 1,\ 8,\ 27, \qquad M = n^3.$

Across M primitive cells, the charge per cell is approximately

$$\boxed{\text{charge per cell} \sim \frac{\delta}{M}}$$

$$M \uparrow \ \Rightarrow \ \frac{\delta}{M} \to 0.$$

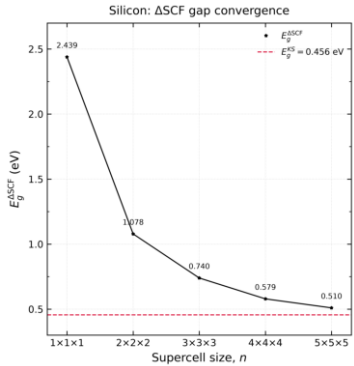
$$E_M(N+\delta) = ME\left(N + \frac{\delta}{M}\right)$$

$$\longrightarrow ME(N) + \delta \left.\frac{\partial E}{\partial N}\right|_{N^+}.$$

Thus the visible curvature becomes progressively smaller.

Important: the curve becoming straighter does not mean that PBEsol becomes exact.
It appears linear because the fractional charge is delocalized over more primitive cells, the bulk limit manifestation of the **delocalization error**.

Si: integer ΔSCF gap versus supercell size



$$E_g^{\Delta\mathrm{SCF}} = E(N-1) - 2E(N) + E(N+1)$$

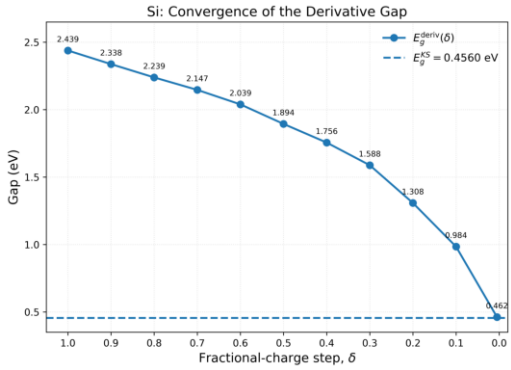
Cell	Atoms	Gap (eV)
$1 \times 1 \times 1$	2	2.4386
$2 \times 2 \times 2$	16	1.0785
$3 \times 3 \times 3$	54	0.7396
$4 \times 4 \times 4$	128	0.5792
$5 \times 5 \times 5$	250	0.5099

Primitive KS reference: $E_g^{KS} = 0.45602719$ eV. At $5 \times 5 \times 5$, the difference is 0.05383781 eV.

Observation

The large decrease with cell size shows that the primitive-cell integer result is not representative of the large-cell limit.

Silicon: Derivative gap approaches the KS gap

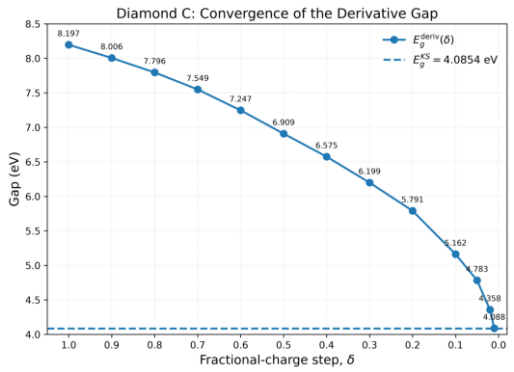


δ	E_g^{deriv} (eV)
1.000	**2.438585**
0.900	2.337560
0.800	2.239009
0.700	2.146623
0.600	2.039067
0.500	1.894242
0.400	1.756080
0.300	1.587980
0.200	1.308105
0.100	0.984380
0.005	0.462000

$$E_g^{deriv}(1) = E_g^{\Delta\mathrm{SCF}} = 2.438585\ \mathrm{eV}.$$

As $\delta \to 0$, the derivative gap approaches the KS-gap scale.

Diamond:Derivative gap approaches the KS gap

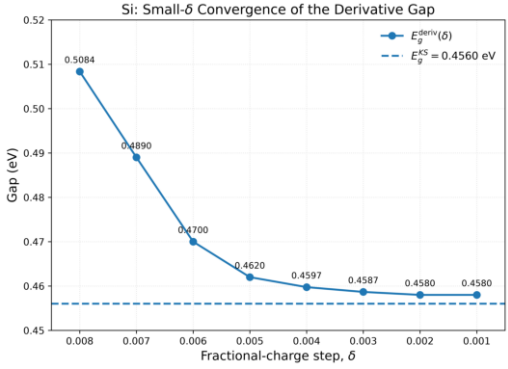


δ	E_g^{deriv} (eV)
0.01	4.087900
0.02	4.357650
0.05	4.783020
0.10	5.162420
0.20	5.790635
0.30	6.199200
0.40	6.575005
0.50	6.908788
0.60	7.247048
0.70	7.549423
0.80	7.795967
0.90	8.005744
1.00	**8.197375**

$E_g^{KS} = 4.0854\ \mathrm{eV}$

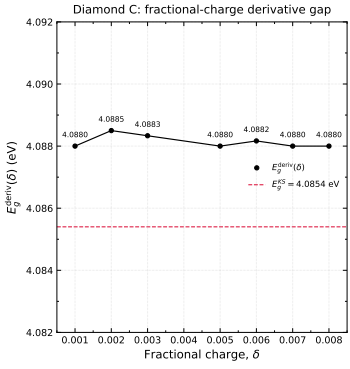
small-δ derivative convergence

Silicon



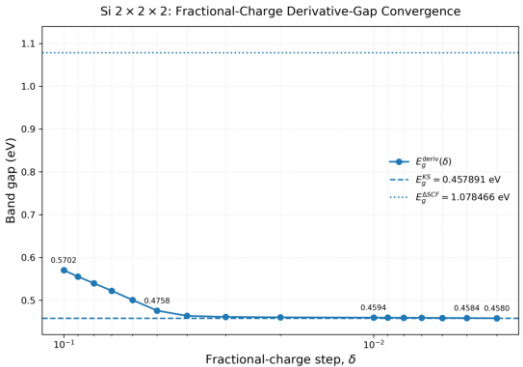
$E_g^{deriv}(0.001) = \mathbf{0.4580}$ eV
Difference from KS: **2 meV**

Diamond



$E_g^{deriv}(0.001) = \mathbf{4.0880}$ eV
Difference from KS: ≈ 2.6 **meV**

$2 \times 2 \times 2$ Si: fractional-charge convergence



For this supercell, $N = 64$.

δ	E_g^{deriv} (eV)
0.100	0.570220
0.050	0.475820
0.010	0.459400
0.008	0.459125
0.006	0.458667
0.005	0.458400
0.004	0.458000

$$E_g^{KS} = 0.45789129 \text{ eV}$$

$$E_g^{deriv}(0.004) - E_g^{KS} = 0.10871 \text{ meV}$$

Key result

At $\delta = 0.004$, the derivative-gap estimate reproduces the same-cell KS gap to within approximately 0.109 meV.

Three gap comparison

Material	E_g^{KS}	E_g^{deriv}	$E_g^{integer}$
Si	0.456027	0.458000	2.438585
Diamond	4.0854	$\approx$ 4.0880	8.197374